3D Stories: Bringing Cultural Heritage Objects to Life

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Figure 1: Rendered historic garments with the 3D Stories logo, showcasing the visual richness and diversity of the collection.

Introduction

The " 3D Stories " project is a collaborative initiative between the Centre for Contemporary and Digital History (C²DH) at the University of Luxembourg and the Urban Complexity Lab (UCLAB) at Potsdam University of Applied Sciences . This partnership has resulted in an innovative interface designed to let users explore and learn about historic garments through interactive 3D models and narrative-driven experiences. By combining detailed notes with smooth storytelling, " 3D Stories " brings history to life in a way that is both engaging and educational.

Challenges

Digitizing and presenting historic garments poses significant challenges. Textiles as cultural artifacts are difficult to represent digitally due to their intricate details and textures. Additionally, platforms like Sketchfab are transitioning to a paid model in 2025, raising concerns about accessibility and the sustainability of digital heritage projects. The large size of 3D scanned files also makes them incompatible with many mobile devices, further limiting access. There is also a lack of a narrative framework in existing platforms that could contextualize these objects meaningfully for diverse audiences.

Solutions

" 3D Stories " addresses these challenges by developing a multi-platform interface that allows users to interact with 3D models through guided narratives or independent exploration. It aims to preserve 3D assets in a sustainable, open-access way while making them accessible across devices. By integrating narrative elements, the tool creates a richer experience and encourages deeper understanding of historical garments. Future versions aim to automate storytelling and animation to further enhance the user experience.

Implementation

The interface allows different approaches to exploring the 3D models. Users can either lean back and follow the narrative or use the 'Explore' button to roam the 3D model on their terms. This flexibility ensures that " 3D Stories " caters to a wide range of users, from those who prefer a structured storytelling approach to those who enjoy a more interactive and exploratory experience.

Conclusion

" 3D Stories " exemplifies how digital tools can enhance the accessibility and engagement of cultural heritage. It combines 3D modeling with storytelling to democratize historical knowledge and preserve it for future generations. The project serves as a model for future digital heritage initiatives that seek to make complex artifacts like historical garments more accessible, engaging, and contextually rich.

Bibliography

The “3D Stories” project website. Available at: Link

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